

Work Breakdown Structure

Levels 1 to 5

Project: Canalside Regency Apartments

Location: Ridgefield Street, Failsworth, Manchester

Employer: Canalside Regency Developments

Document title: Work Breakdown Structure — Levels 1 to 5

Document number: P5RS_CSRA_01_WBS_ISO_MASTER

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This is a controlled document. It defines project scope only and does not constitute a programme, cost plan, commercial valuation, procurement tracker or evidence of appointment.

Document Control

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Checked by	
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Document Purpose

This Work Breakdown Structure defines the baseline scope of the Works for the Canalside Regency Apartments development.

It identifies the project-management, professional-design, procurement, construction, testing, certification and handover deliverables required for completion of the development.

The WBS is a scope-definition document. It records scope inclusions and deliverables and establishes the coding framework by which they are organised.

The WBS is not

- A programme, method statement, cost plan or commercial valuation.
- A procurement-status tracker or a risk register.
- Evidence that a consultant or subcontractor has been appointed.

Scope completeness and drawing control

The WBS shall be read with the current approved Employer's Requirements, Contractor's Proposals, drawings, specifications, schedules, statutory approvals and planning-condition information. Historic drawings are reference information only until their current status and contract applicability are confirmed.

No work item is excluded solely because it is not repeated under every interfacing package. Any proposed exclusion, employer-supplied item or work by others shall be stated in the relevant procurement package and approved before award.

The contract boundary, completed works, retained existing assets, third-party interfaces and any works outside the site boundary shall be confirmed against the current approved site and external-works information.

Project-specific scope baseline

The development comprises a single residential building of part four, five and six storeys providing 138 apartments, comprising 96 two-bedroom apartments and 42 one-bedroom apartments.

The building includes two stair cores, two passenger lifts, central communal circulation, apartment balconies, reception facilities, a parcel store, a communal WC and associated communal accommodation.

Communal accommodation of approximately 37 m² is provided at each applicable floor, subject to verification against the current approved architectural general-arrangement drawings and Schedule of Accommodation.

The external works comprise roads, parking, accessible parking, electric-vehicle charging provision, cycle storage, bin stores, paving, drainage, external lighting, boundary treatments, private gardens, soft landscaping, canal-side interfaces and works outside the site boundary where required by approved planning or third-party information.

The project scope shall be reconciled against the current approved red-line boundary and shall distinguish the apartment building, apartment external works, shared infrastructure, completed or separately constructed housing plots, private drives and gardens, third-party works, employer-supplied items, work by others and retained existing assets.

No historic site, landscape or boundary drawing shall define the contract scope unless its status and applicability have been formally confirmed.

Project-specific exclusions and optional items

Unless expressly included within the approved Employer's Requirements, Contractor's Proposals, drawings or specification, window blinds, fitted wardrobes, mirrors, fixed bathroom accessories, purchaser upgrade packs and loose decorative furniture are excluded or optional.

Each trade-package scope shall identify all exclusions, employer-supplied items and work by others before subcontract award.

WBS level definitions

Level	Definition	Example
Level 1	Project stage	4.0 Construction
Level 2	Professional workstream, trade or building system	4.3 Substructure
Level 3	Deliverable element	4.3.6 Retaining walls
Level 4	Building zone	Zone A · Core · External
Level 5	Floor or storey	A.00 — Zone A, ground floor

Levels 4 and 5 form a location-coding overlay applied to Level 3 deliverables; the coding protocol and styles are defined in the annexes so the breakdown can be expanded by zone and floor without rebuilding the numbering system.

1.0 PROJECT MANAGEMENT AND GOVERNANCE

1.1 Project Management

1.1.1 Project governance and administration

- (a) Establish and maintain the project governance structure.
- (b) Maintain the project directory.
- (c) Establish reporting and approval routes.
- (d) Coordinate the employer, Main Contractor, consultants, specialists and trade contractors.
- (e) Manage project meetings, actions and decisions.
- (f) Maintain project risk, change and decision records.
- (g) Escalate unresolved matters affecting design, procurement, construction or completion.
- (h) Coordinate project reporting through to handover.

1.1.2 Project programme coordination

- (a) Coordinate design deliverables with the project programme.
- (b) Coordinate statutory approvals with procurement and construction.
- (c) Identify information required for trade-package procurement.
- (d) Identify information required for construction release.
- (e) Monitor design and approval dependencies.
- (f) Coordinate commissioning, certification and handover requirements with the completion programme.

1.1.3 Stakeholder coordination

- (a) Employer and Employer's Agent coordination.
- (b) Consultant coordination.
- (c) Building Control coordination.
- (d) Structural warranty-provider coordination.
- (e) Utility-provider coordination.
- (f) Planning-authority coordination.
- (g) Canal-side and third-party interface coordination.
- (h) Neighbour and local stakeholder coordination where required.
- (i) Relevant canal authority, canal-side landowner and third-party land-interest coordination.
- (j) ENWL, appointed ICP, United Utilities, sewerage undertaker, telecommunications provider and highway-authority coordination.
- (k) Coordination of licences, consents, easements, wayleaves, leases and adoption agreements required for construction, connection, occupation and future maintenance.
- (l) Maintain a third-party approvals schedule identifying the responsible party, required submission, approval status and completion evidence.

1.2 Design Management

1.2.1 Design delivery management

- (a) Establish and maintain the design programme.
- (b) Establish and maintain the design deliverables schedule.
- (c) Coordinate Stage 4 and construction information.
- (d) Coordinate consultant and specialist-subcontractor design.
- (e) Manage interdisciplinary design reviews.
- (f) Coordinate buildability reviews.
- (g) Coordinate design information required for procurement.
- (h) Manage RFIs and technical queries.
- (i) Monitor design approvals and information status.
- (j) Coordinate design information required for testing and handover.

1.2.2 Design responsibility management

- (a) Establish and maintain the Design Responsibility Matrix.
- (b) Define consultant design responsibilities.
- (c) Define specialist and trade-contractor design responsibilities.
- (d) Identify Contractor's Design Portions.
- (e) Identify interfaces between trade packages.
- (f) Identify delegated-design and performance-specified elements.
- (g) Coordinate design-review and acceptance requirements.
- (h) Prevent gaps or duplication between appointments and trade scopes.
- (i) Prepare and maintain a project-specific scope and interface matrix covering design, supply, installation, coordination, testing, certification and handover responsibilities.
- (j) Allocate each interface to a named consultant, specialist or trade contractor.
- (k) Record all exclusions, work by others, employer-supplied items and third-party works.
- (l) Confirm the project location-coding structure against the approved architectural general-arrangement drawings and replace generic zone references where they do not correspond with the approved building arrangement.

1.2.3 Design change control

- (a) Record proposed design changes.
- (b) Assess technical, statutory, commercial and programme implications.
- (c) Obtain the required approvals before implementation.
- (d) Coordinate revised drawings, specifications, calculations and schedules.
- (e) Withdraw superseded information.
- (f) Communicate approved changes to affected parties.
- (g) Maintain the Golden Thread of design decisions and information.

1.3 Information Management

1.3.1 Common Data Environment and document control

- (a) Administer the Common Data Environment.
- (b) Manage document naming, numbering, revision and suitability status.
- (c) Manage review and approval workflows.
- (d) Maintain document and drawing registers.
- (e) Maintain information-delivery schedules.
- (f) Control superseded information.
- (g) Coordinate published, shared, work-in-progress and archived information.
- (h) Maintain information required for the Golden Thread.

1.3.2 BIM and digital coordination

- (a) Establish BIM and information-management requirements.
- (b) Coordinate model exchanges.
- (c) Undertake federated-model reviews.
- (d) Coordinate spatial, structural and M&E models.
- (e) Record and close coordination issues.
- (f) Coordinate record models and asset information where required.

1.3.3 Information Management System

- (a) Define the project IMS requirements.
- (b) Establish information-management procedures.
- (c) Maintain secure access and information-control requirements.
- (d) Coordinate digital design, construction and handover records.
- (e) Coordinate final asset and record information.

1.4 Health, Safety and Design Risk

1.4.1 CDM compliance

- (a) Coordinate the Principal Designer requirements under CDM.
- (b) Maintain the design-risk register.
- (c) Coordinate design-risk reviews.
- (d) Record significant residual risks.
- (e) Coordinate pre-construction information.
- (f) Communicate residual risks to designers and contractors.
- (g) Coordinate information required for the Health and Safety File.

1.4.2 Construction health and safety management

- (a) Prepare and maintain the Construction Phase Plan.
- (b) Manage site health and safety.
- (c) Coordinate permits and inspection records.
- (d) Manage temporary-works risks.
- (e) Maintain emergency arrangements.
- (f) Maintain temporary site fire precautions.
- (g) Monitor contractor compliance.
- (h) Maintain health-surveillance requirements.

1.5 Commercial and Contract Administration

1.5.1 Contractor commercial management

- (a) Package cost management.
- (b) Procurement comparisons.
- (c) Change control.
- (d) Variation management.
- (e) Valuation and payment assessment.
- (f) Forecasting.
- (g) Subcontract account management.
- (h) Final-account management.

1.5.2 Employer's Agent and Contract Administrator interface

- (a) Coordinate instructions and technical responses.
- (b) Coordinate valuation and payment information.
- (c) Coordinate change information.
- (d) Coordinate inspections.
- (e) Coordinate Practical Completion requirements.
- (f) Coordinate defect and close-out requirements.
- (g) Maintain supporting contract records.

1.5.3 Project insurance

- (a) Contractor's all-risk insurance.
- (b) Cover for the Works and project materials.
- (c) Cover for temporary works where required.
- (d) Cover for existing structures where applicable.
- (e) Consultant and subcontractor insurance requirements.
- (f) Professional indemnity insurance evidence.
- (g) Policy records and renewal requirements.

2.0 DETAILED DESIGN AND PROFESSIONAL SERVICES

2.1 Lead Architect

2.1.1 Architectural design

- (a) Stage 4 and construction architectural information.
- (b) General arrangement plans.
- (c) Elevations and sections.
- (d) Apartment layouts.
- (e) Communal-area layouts.
- (f) External-wall and façade details.
- (g) Internal-construction details.
- (h) Roof, parapet and balcony details.
- (i) Door, window and ironmongery coordination.
- (j) Finishes and room-data information.
- (k) Kitchen, bathroom and FF&E coordination.
- (l) Final Schedule of Accommodation covering apartments, communal areas, reception, parcel store, communal WC, plant and service spaces.
- (m) Reception, parcel-store and communal-WC layouts, finishes, accessibility, security, fire-safety and building-services coordination.
- (n) Communal accommodation at each applicable floor coordinated against the approved use and general-arrangement drawings.
- (o) Dwelling-accessibility schedule identifying each apartment as M4(1), M4(2) or M4(3), where applicable.
- (p) Accessible entrances, thresholds, circulation, doors, sanitary facilities, controls and parking interfaces.
- (q) Façade and plantroom louvre locations, dimensions, appearance, weathering, acoustic performance and building-envelope interfaces.
- (r) Apartment mailboxes, parcel-management arrangements, reception furniture, signage, access control, CCTV, power and data requirements.
- (s) Accessibility coordination.
- (t) Planning and Building Regulations coordination.
- (u) Coordination with structural, civil and M&E information.

2.1.2 Architectural construction support

- (a) Respond to RFIs.
- (b) Review specialist-subcontractor information.
- (c) Review material and product submissions.
- (d) Review samples and mock-ups.
- (e) Undertake inspections where included within the appointment.
- (f) Coordinate architectural defects and remedial details.
- (g) Prepare record and as-built information.

2.2 Structural and Civil Engineer

2.2.1 Structural design

- (a) Piling design criteria and pile coordination.
- (b) Pile caps and ground beams.
- (c) Ground-floor structural system.
- (d) Structural-steel frame.
- (e) Stability and bracing systems.
- (f) Suspended structural floors.
- (g) Precast staircases.
- (h) Balcony structural interfaces.
- (i) Lift-shaft framing.

- (j) Structural openings.
- (k) Structural robustness.
- (l) Structural movement joints.
- (m) Retaining walls.
- (n) Structural bases and plinths.
- (o) Sleeper walls and masonry support walls.
- (p) Oversite concrete and ground-floor build-up interfaces.
- (q) Structural-steel fire-protection requirements and connection details.
- (r) Structural calculations and specifications.

2.2.2 Civil-engineering design

- (a) Site levels and earthworks.
- (b) Below-ground foul drainage.
- (c) Below-ground surface-water drainage.
- (d) Attenuation.
- (e) Drainage chambers and manholes.
- (f) External service routes.
- (g) Roads and parking.
- (h) Paving and kerbs.
- (i) Retaining-wall drainage.
- (j) Utility coordination.
- (k) External levels and thresholds.
- (l) Landscaping interfaces.
- (m) Private-garden layouts, levels, drainage and boundary interfaces.
- (n) Tree pits, root barriers and below-ground service clearances.
- (o) Landscape works outside the site boundary where required by the approved planning information.
- (p) Section 278, highway and third-party reinstatement requirements where applicable.
- (q) Current red-line contract boundary and schedule of shared, third-party and off-site works.
- (r) Final external-works scope drawing and schedule covering roads, parking, private drives, paving, kerbs, drainage, lighting, boundaries, gates, private gardens and landscaping.
- (s) Defensive Planting Scope Drawing referenced by the boundary-treatment information.
- (t) Final planting schedule, planting specification and landscape-establishment requirements.
- (u) Land ownership, access rights and future maintenance responsibilities for landscaping outside the site boundary.
- (v) Canal-side levels, retaining structures, drainage, outfalls, access and protection requirements coordinated with the relevant authority or landowner.
- (w) Highway agreements under Sections 38 and 278 where applicable.
- (x) Sewer connection, adoption and diversion agreements under Sections 104, 106 and 185 where applicable.
- (y) Utility easements, wayleaves, substation leases and access requirements.
- (z) Landscape maintenance and establishment period required by the approved planning and landscape information.

2.2.3 Structural and civil construction support

- (a) Review specialist-design submissions.
- (b) Respond to RFIs.
- (c) Review temporary-condition interfaces.
- (d) Review remedial proposals.
- (e) Undertake inspections where included within the appointment.
- (f) Provide record drawings, calculations and completion information.

2.3 Mechanical and Electrical Engineer

2.3.1 Mechanical design

- (a) Incoming water supply.
- (b) Water storage and booster systems.
- (c) Domestic hot- and cold-water distribution.

- (d) Water metering.
- (e) Sanitaryware connections.
- (f) Above-ground foul drainage and soil vent pipes.
- (g) Wastewater heat recovery where applicable.
- (h) Apartment and communal ventilation.
- (i) Apartment space-heating and cooling strategy.
- (j) Heating emitters or indoor units, external units, controls, condensate and maintenance access.
- (k) Acoustic, vibration, façade, balcony, fire-stopping and weathering requirements for heating and cooling equipment.
- (l) External-bin-store foul drainage, floor falls, washdown and ventilation requirements.
- (m) Plantroom ventilation.
- (n) Smoke-control systems.
- (o) AOV systems.
- (p) Sprinkler systems.
- (q) Dry risers.
- (r) Mechanical controls.
- (s) Mechanical builders' work requirements.
- (t) Mechanical commissioning requirements.
- (u) Wastewater heat-recovery design where required by the approved SAP and energy strategy.
- (v) Wastewater heat-recovery units coordinated with bath wastes, drainage falls, access panels, structure, acoustic treatment and maintenance access.
- (w) Water, drainage and ventilation services to the reception, parcel store, communal WC and communal spaces.
- (x) Plantroom and pump-room layout, including tanks, booster pumps, drainage, ventilation, access, replacement routes, bases, lighting, power and fire-stopping requirements.
- (y) Mechanical requirements for façade and plantroom intake and extract louvres, including free area, pressure drop, weather, acoustic, mesh and access requirements.
- (z) Separate domestic and sprinkler incoming water services coordinated with United Utilities and the relevant connection provider.
- (aa) Current project requirements of a 90 mm barrier-pipe domestic supply designed for 6 l/s and a 63 mm barrier-pipe sprinkler supply designed for 3 l/s, subject to final provider approval and coordinated hydraulic design.

2.3.2 Electrical design

- (a) Incoming electrical supply.
- (b) DNO substation interfaces.
- (c) DNO/ICP earthing study and approval.
- (d) Existing electricity and telecommunications diversion requirements.
- (e) Electrical intake and main distribution.
- (f) Landlord and apartment distribution.
- (g) Containment.
- (h) Metering.
- (i) Consumer units.
- (j) Small power.
- (k) Internal and external lighting.
- (l) Emergency lighting.
- (m) Heating and appliance supplies.
- (n) Data, telephone, TV and satellite systems.
- (o) Fire alarm.
- (p) CCTV.
- (q) Access control and door entry.
- (r) Electric-vehicle charging, load management, metering, communications and revenue-control strategy.
- (s) Lightning and surge protection.
- (t) Solar PV and SolShare.
- (u) Life-safety power supplies.

- (v) Standby generator and automatic changeover.
- (w) Electrical builders' work requirements.
- (x) Electrical testing and commissioning requirements.
- (y) Division between ENWL non-contestable works and ICP contestable works.
- (z) Accepted ENWL Option 2 point of connection coordinated with the appointed ICP design and construction scope.
- (aa) Responsibility for substation construction, DNO equipment, incoming and outgoing cables, ducts, chambers, earthing, metering, energisation and reinstatement.
- (bb) DNO substation lease, access requirements, easements and wayleaves.
- (cc) Confirmed electric-vehicle charger quantity, locations, active and passive provision, load management, metering, communications, payment and revenue-management strategy.
- (dd) Electrical services to the reception, parcel store, communal WC and communal spaces.

2.3.3 M&E coordination

- (a) Plant-space requirements.
- (b) Riser and ceiling-zone coordination.
- (c) Service routes.
- (d) Maintenance and replacement access.
- (e) Equipment schedules.
- (f) Builders' work schedules.
- (g) Penetration and fire-stopping schedules.
- (h) Power requirements for mechanical equipment.
- (i) Controls and cause-and-effect interfaces.
- (j) Acoustic and vibration requirements.
- (k) Commissioning and handover information.

2.4 Fire Engineer

2.4.1 Fire strategy

- (a) Means of escape.
- (b) Occupancy assumptions.
- (c) Compartmentation.
- (d) Fire-resistance requirements.
- (e) Structural-steel fire-protection performance, inspection and certification requirements.
- (f) Cavity barriers.
- (g) External-wall fire performance.
- (h) Fire doors.
- (i) Fire alarm and detection.
- (j) Sprinkler requirements.
- (k) Dry-riser requirements.
- (l) Smoke-control strategy.
- (m) AOV requirements.
- (n) Firefighting access and facilities.
- (o) Life-safety and standby-power requirements.
- (p) Fire-stopping principles.
- (q) Cause-and-effect requirements.
- (r) Regulation 38 information.

2.4.2 Fire-design coordination

- (a) Coordinate architectural, structural and M&E fire requirements.
- (b) Review specialist fire-system designs.
- (c) Review fire-stopping proposals.
- (d) Review façade fire-performance information.
- (e) Coordinate fire-strategy changes.
- (f) Support Building Control submissions.

- (g) Support testing, commissioning and handover.

2.5 Energy Assessor

2.5.1 Energy and Building Regulations compliance

- (a) Design-stage SAP calculations.
- (b) Part L compliance.
- (c) Thermal-envelope performance.
- (d) Thermal-bridge coordination.
- (e) Air-permeability requirements.
- (f) Heating, hot-water and ventilation assumptions.
- (g) Renewable-energy assumptions.
- (h) Installed-product performance data.
- (i) As-built SAP calculations.
- (j) Energy Performance Certificates.
- (k) Design-stage overheating assessment in accordance with Approved Document O.
- (l) CIBSE TM59 modelling where required by the approved compliance strategy, planning requirements or Employer's Requirements.
- (m) Coordination of glazing areas, glazing g-values, solar gain, shading, purge ventilation, background ventilation, internal gains and cooling assumptions.
- (n) Overheating-mitigation measures recorded on the coordinated architectural and M&E design information.
- (o) Minimum EPC rating B target for each apartment, subject to the approved energy strategy and final as-built assessment.

2.6 Acoustic Consultant

2.6.1 Acoustic design and testing

- (a) Part E design requirements.
- (b) Apartment-separation performance.
- (c) Wall and floor acoustic build-ups.
- (d) Flanking-detail review.
- (e) Façade acoustic requirements.
- (f) Plant and ventilation noise criteria.
- (g) Construction-stage acoustic advice.
- (h) Pre-completion acoustic testing.
- (i) Final acoustic report.

2.7 Principal Designer — CDM

2.7.1 CDM Principal Designer services

- (a) Plan, manage and monitor the pre-construction phase.
- (b) Coordinate design health and safety.
- (c) Review designer compliance.
- (d) Coordinate the design-risk register.
- (e) Coordinate pre-construction information.
- (f) Coordinate significant residual risks.
- (g) Liaise with the Principal Contractor.
- (h) Coordinate information for the Health and Safety File.

2.8 Building Regulations Principal Designer

2.8.1 Building Regulations design management

- (a) Plan, manage and monitor design work for Building Regulations compliance.
- (b) Coordinate designer competence and compliance.
- (c) Coordinate Building Regulations submissions.

- (d) Coordinate changes affecting compliance.
- (e) Maintain compliance records.
- (f) Coordinate completion declarations and dutyholder information.

2.9 Building Control

2.9.1 Building Control approval

- (a) Building Regulations application.
- (b) Design submissions and responses.
- (c) Plan-review comments.
- (d) Inspection schedule.
- (e) Site inspections.
- (f) Design-change notifications.
- (g) Compliance evidence.
- (h) Final inspection.
- (i) Completion documentation.
- (j) Building Control Final Certificate.

2.10 Structural Warranty

2.10.1 Structural warranty requirements

- (a) Development registration.
- (b) Technical-audit submissions.
- (c) Responses to technical-audit comments.
- (d) Staged inspections.
- (e) Design and construction evidence.
- (f) Closure of warranty observations.
- (g) Warranty cover note.
- (h) Final warranty certificates and policy documents.

2.11 Water-Efficiency Assessment

2.11.1 Part G calculations

- (a) Establish fitting flow rates and capacities.
- (b) Coordinate sanitaryware and appliance assumptions.
- (c) Prepare design-stage calculations.
- (d) Update calculations using installed products.
- (e) Produce final Part G compliance calculations.

2.12 Quantity Surveyor and Cost Consultant

2.12.1 Cost-management services

- (a) Cost-plan management.
- (b) Package budgets.
- (c) Procurement support.
- (d) Change assessment.
- (e) Cost-risk monitoring.
- (f) Valuation review.
- (g) Forecasting.
- (h) Final-account support.

2.13 Employer's Agent

2.13.1 Contract-administration services

- (a) Administration of the Building Contract.
- (b) Issue of contractual instructions, notices and certificates.
- (c) Review of payment applications and variations.
- (d) Review of delay and extension-of-time submissions.
- (e) Administration of Practical Completion.
- (f) Administration of the defects period and final completion.
- (g) Coordination of Employer decisions and approvals.
- (h) Maintenance of contractual records.

2.14 Clerk of Works

2.14.1 Independent quality inspections

- (a) Independent inspection of workmanship and materials.
- (b) Inspection of samples, benchmarks and completed installations.
- (c) Site inspection reports and photographic records.
- (d) Identification and reporting of defective or non-compliant work.
- (e) Follow-up inspections and closure of observations.
- (f) Main Contractor access, attendance and information provision.
- (g) Inspection and reporting only, with no authority to instruct changes to the Works.

2.15 Independent Monitoring Surveyor

2.15.1 Independent project monitoring

- (a) Independent monitoring on behalf of the Employer or funder.
- (b) Review of construction progress and programme status.
- (c) Review of quality, risks and statutory approvals.
- (d) Review of valuations and drawdown information where required.
- (e) Periodic site inspections and monitoring reports.
- (f) Review of completion, certification and handover status.
- (g) Main Contractor access, attendance and information provision.
- (h) Monitoring and reporting only, with no authority to instruct changes to the Works.

3.0 PROCUREMENT AND TRADE PACKAGES

3.1 Preliminaries and Site-Wide Packages

3.1.1 Preliminaries

- (a) Trade-package scope schedules, drawings, specifications and interface matrices.
- (b) Tender issue, tender queries, bid comparison and scope-gap closure.
- (c) Subcontract award, contract execution, design obligations, insurances, bonds and warranties where required.
- (d) Long-lead procurement, samples, technical submittals and approval requirements.

3.1.2 Site services and general attendance

- (a) Temporary water for construction, welfare, cleaning, dust suppression, testing and commissioning.
- (b) Temporary water connections, tanks, pumps, distribution pipework, hoses, standpipes and frost protection.
- (c) Temporary electricity supplies, distribution boards, transformers, cabling, sockets and task lighting.
- (d) Temporary heating, drying, dehumidification and ventilation required to maintain suitable construction conditions.
- (e) Site heaters required for curing, frost protection, drying, decoration, flooring, commissioning and protection of completed works.
- (f) Fuel, electricity, water and other temporary-service consumption charges.
- (g) Temporary drainage, pumping, discharge, lighting, telecommunications and internet arrangements.
- (h) Inspection, testing, maintenance, relocation, extension, removal and making good of temporary services.
- (i) General attendance to subcontractors, including shared access, unloading areas, temporary services, waste facilities and welfare.
- (j) The package scope shall define which temporary services and attendances are provided by the Main Contractor and which are included by each subcontractor.

3.1.3 Scaffolding and access

3.1.4 Lifting solutions and site plant

3.1.5 Temporary works

3.1.6 Waste and muck-away services

3.1.7 Testing and commissioning specialists

3.1.8 Handover documentation and O&M coordination

3.1.9 Package scope and interface control

- (a) Complete supply, installation and Contractor's Design Portion responsibilities where applicable.
- (b) All labour, supervision, access, plant, fixings, supports, sundries and consumables required for a complete installation.
- (c) Package drawings, calculations, specifications, schedules and builders' work information.
- (d) Coordination with preceding, adjacent and following trades.
- (e) Defined inclusions, exclusions, employer-supplied items and work by others.
- (f) Technical submissions, samples, benchmarks and mock-ups.
- (g) Consultant, Building Control, warranty-provider and statutory-authority approvals where applicable.
- (h) Protection, temporary works, attendance, making good and final cleaning.
- (i) Inspection, testing, commissioning and certification.
- (j) As-built drawings, O&M information, asset information, warranties and training.
- (k) Compliance with the project scope and interface matrix and the separate procurement, design and construction programmes.
- (l) Closure of all qualifications and scope gaps before subcontract award.

3.2 Enabling Works and Substructure Packages

3.2.1 Enabling-works contractor

3.2.2 Remediation and invasive-weed treatment contractor

3.2.3 Existing-service protection and diversion contractor

3.2.4 Groundworks contractor

3.2.5 Piling contractor

3.2.6 Pile-cap and ground-beam contractor

3.2.7 Retaining-wall works

3.2.8 Below-ground drainage contractor

3.2.9 Multi-utility contractor

- (a) ENWL non-contestable works and ICP contestable works coordination.
- (b) United Utilities domestic and sprinkler-water connections.
- (c) Sewer connection and adoption requirements.
- (d) Telecommunications and FFTP connections.
- (e) Ducts, chambers, draw pits, draw cords, markers, service entries, testing and record information included within the package.
- (f) Definition of utility-provider works, contractor works, groundworker attendance and reinstatement responsibilities.
- (g) Substation leases, easements, wayleaves, access and adoption requirements.

3.2.10 Ground-floor structural system

3.3 Superstructure Packages

3.3.1 Structural-steel-frame contractor

3.3.2 Structural-steel fire-protection contractor

3.3.3 Suspended structural-floor contractor

3.3.4 Precast-stair contractor

3.3.5 Structural grouting, ties and infill works

3.3.6 Balcony structural works

3.4 Building-Envelope Packages

3.4.1 SFS and cladding contractor

3.4.2 Masonry contractor

3.4.3 Windows, external doors and curtain-wall contractor

- (a) Façade and entrance louvres required by the coordinated architectural and M&E design.
- (b) Louvre frames, blades, supports, weathering, drainage, acoustic treatment, bird and insect mesh, fire performance and perimeter sealing.
- (c) Coordination of louvre openings and interfaces with the façade, SFS, masonry, waterproofing and M&E installations.

3.4.4 Roofing contractor

3.4.5 Balcony-decking installer

3.4.6 Architectural metalwork and balustrade contractor

3.4.7 Mastic and sealant contractor

3.4.8 Passive-fire and cavity-barrier contractor

3.5 Internal Fit-Out Packages

3.5.1 Drylining contractor

3.5.2 Ceiling-systems contractor

3.5.3 Plastering contractor

3.5.4 Internal doors and ironmongery contractor

3.5.5 General-joinery contractor

3.5.6 Screeding contractor

3.5.7 Flooring contractor

3.5.8 Tiling contractor

3.5.9 Painting and decorating contractor

3.5.10 Kitchen contractor

3.5.11 Bathroom fit-out and sanitaryware contractor

3.5.12 FF&E contractor

3.5.13 Signage and wayfinding contractor

3.5.14 Reception, parcel-store, communal-WC and communal-area fit-out package

- (a) Reception, entrance, parcel-store, communal-WC and communal-area fit-out.
- (b) Finishes, joinery, mailboxes, storage, shelving or parcel lockers where scheduled.
- (c) Doors, ironmongery, access control, CCTV, lighting, power, data, fire-alarm, water, drainage and ventilation interfaces.
- (d) Accessibility, testing, protection, cleaning and handover requirements.

3.6 Building-Services Packages

3.6.1 Mechanical building-services contractor

- (a) Wastewater heat-recovery installation where required by the approved energy strategy.
- (b) Mechanical services to the reception, parcel store, communal WC and communal spaces.
- (c) Complete plantroom and pump-room installation and fit-out.
- (d) Intake and extract louvre coordination with the façade contractor.
- (e) Final connections, testing, commissioning, certification and O&M information.

3.6.2 Apartment heating and cooling systems contractor

3.6.3 Electrical-services contractor

3.6.4 Electric-vehicle charging contractor

3.6.5 Plantroom-installation contractor

3.6.6 Ventilation contractor

3.6.7 Smoke-control and AOV specialist

3.6.8 Dry-riser installer

3.6.9 Sprinkler-system contractor

3.6.10 Fire-alarm installer

3.6.11 Security-systems contractor

3.6.12 Solar PV and SolShare contractor**3.6.13 Lift contractor****3.6.14 Standby-generator specialist****3.6.15 Permanent fire-extinguisher supplier****3.6.16 Builders' work in connection****3.7 External-Works Packages****3.7.1 Roads, parking and paving contractor****3.7.2 Section 278 and highway-works contractor where required****3.7.3 Cycle-store and bin-store contractor****3.7.4 External drainage contractor****3.7.5 Landscaping contractor****3.7.6 Fencing and boundary-treatment contractor****3.7.7 External signage and street-furniture contractor****3.7.8 Canal-side and third-party interface works**

- (a) Canal-side access, protection and temporary works.
- (b) Retaining-wall, boundary, drainage and outfall interfaces.
- (c) Third-party land access, licences, protection and making good.
- (d) Relevant authority or landowner inspections, approvals, completion records and future-maintenance information.

4.0 CONSTRUCTION

4.1 Preliminaries

4.1.1 Site establishment

- (a) Site offices and cabins.
- (b) Canteen and rest facilities.
- (c) Drying and changing facilities.
- (d) Site-accommodation furniture.
- (e) Hoardings and fencing.
- (f) Site signage.
- (g) Initial site survey and setting out.
- (h) Site-access control.
- (i) Compound, laydown and storage areas.

4.1.2 Temporary services

- (a) Temporary incoming water supply, including water for construction, welfare, cleaning, dust suppression, flushing, pressure testing, chlorination and commissioning.
- (b) Temporary electricity supply, metering, distribution boards, transformers, cabling and socket outlets.
- (c) Temporary drainage, sump pumping, dewatering and approved discharge arrangements.
- (d) Temporary telecommunications and internet.
- (e) Temporary task, access, security and emergency lighting.
- (f) Maintenance and removal of temporary services.
- (g) Temporary heating for frost protection, concrete curing, drying, decoration, flooring, commissioning and protection of completed works.
- (h) Temporary dehumidification, drying, ventilation and air-movement equipment.
- (i) Monitoring of temperature, relative humidity and substrate moisture. Govee system or similar.
- (j) Water, electricity, fuel and other temporary-service consumption charges.
- (k) Removal of temporary installations and making good before completion.

Note DT: - Direct-fired heaters shall not be used within enclosed or substantially completed areas without an approved risk assessment, ventilation arrangement and moisture-control assessment.

4.1.3 Scaffolding and access equipment

- (a) External scaffolding.
- (b) Internal scaffolding.
- (c) Scaffolding stairs tower, Haki system or similar
- (d) Mast climbers.
- (e) Goods and/or passenger hoists.
- (f) Access towers, podiums and platforms.
- (g) Edge protection, fall protection and safety netting.
- (h) Temporary stairs and ramps.
- (i) Loading bays.
- (j) Inspections, adaptations and removal.

4.1.4 Site security

- (a) Security guards and patrols.
- (b) Temporary CCTV.
- (c) Alarm systems.
- (d) Security lighting.
- (e) Secure fencing and gates.
- (f) Visitor and delivery control.

4.1.5 Health and safety provisions

- (a) First-aid facilities.
- (b) Personal protective equipment.
- (c) Safety signage and barriers.
- (d) Temporary extinguishers and alarms.
- (e) CDM administration.
- (f) Permits and inspections.
- (g) Emergency arrangements.
- (h) Health-surveillance monitoring.
- (i) Dust Suppressions
- (j) Construction noise and vibration monitoring where required.

4.1.6 Temporary roads and logistics

- (a) Temporary roads and paths.
- (b) Laydown and storage areas.
- (c) Wheel-washing facilities.
- (d) Traffic marshals and banksmen.
- (e) Delivery scheduling.
- (f) Road cleaning and sweeping.
- (g) Traffic-management measures.
- (h) Material handling and storage.

4.1.7 Site cleaning and waste management

- (a) Cleaning of site accommodation.
- (b) General waste skips.
- (c) Hardcore-waste skips.
- (d) Timber-waste skips.
- (e) Metal-waste skips.
- (f) Plasterboard-waste skips.
- (g) Waste segregation.
- (h) Waste-transfer documentation.
- (i) Final waste removal.

4.1.8 Welfare facilities

- (a) Toilet block.
- (b) Washing facilities.
- (c) Drinking water.
- (d) Rest and changing facilities.
- (e) Welfare cleaning and maintenance.

4.1.9 Site management and general attendance

- (a) Site management.
- (b) Assistant site management.
- (c) Trade and logistics supervision.
- (d) Site labour.
- (e) Plant operation.
- (f) General attendance.
- (g) Site consumables.

4.1.10 General plant

- (a) Tower and mobile cranes.
- (b) Forklifts and telehandlers with buckets, tippers, extra log forks, etc.
- (c) Rough terrain powered pallet trucks.

- (d) Trolleys and dollies for manual handling and material movement.
- (e) General site plant.
- (f) Small tools and equipment.
- (g) Inspection and maintenance.
- (h) Removal following completion.
- (i) Fuel bowser.

4.2 Enabling Works

4.2.1 Site clearance

- (a) Tree-stump and root removal.
- (b) Bush, scrub and undergrowth clearance.
- (c) Small-tree removal.
- (d) Topsoil stripping.
- (e) Stockpiling suitable topsoil.
- (f) Disposal of unsuitable material.
- (g) Arboricultural and ecological controls.

4.2.2 Site grading and earthworks

- (a) Formation levels.
- (b) Cut and fill.
- (c) Movement of material within the site.
- (d) Temporary stockpiles.
- (e) Filling, levelling and compaction.
- (f) Disposal of surplus material.
- (g) Compaction testing.

4.2.3 Surveys and investigations

- (a) Topographical surveys.
- (b) Setting-out surveys.
- (c) Existing services surveys.
- (d) Drainage and condition surveys.

4.2.4 Working platform certification

- (a) Working-platform design.
- (b) Formation preparation.
- (c) Imported material.
- (d) Placement and compaction.
- (e) Testing and certification.
- (f) Maintenance during piling operations.
- (g) Decommissioning, removal, reuse or incorporation of the working platform after piling.
- (h) Regrading, reinstatement and certification of the permanent formation.

4.2.5 Temporary works

- (a) Excavation support and earth-retention systems.
- (b) Temporary stability, propping and bracing.
- (c) Temporary access and working platforms.
- (d) Crane bases, hoist bases and plant foundations.
- (e) Edge protection and loading platforms.
- (f) Temporary drainage, dewatering and sump pumping.
- (g) Temporary-works design, checking and coordination.
- (h) Inspections, permits and monitoring.
- (i) Maintenance, adaptation and removal.
- (j) Making good following removal.

- (k) Compliance with BS 5975.

4.3 Substructure

4.3.1 Excavation and filling

- (a) Verification of existing levels across the building footprint.
- (b) Removal of existing rough hard standing and unsuitable material.
- (c) Reduced-level excavation and local cut-and-fill to approved formation levels.
- (d) Excavation for pile caps and ground beams.
- (e) Excavation for retaining walls.
- (f) Excavation for drainage and services.
- (g) Temporary support to excavations.
- (h) Disposal of surplus material.
- (i) Reuse of suitable excavated material.
- (j) Imported fill.
- (k) Backfilling and layered compaction.
- (l) Compaction testing.

4.3.2 Borehole and pile-probe investigations

- (a) Pile-probe investigations.
- (b) Borehole testing where required.
- (c) Obstruction identification.
- (d) Records and reporting.
- (e) Coordination with the piling design.

4.3.3 CFA piling

- (a) Pile design and setting out.
- (b) Pile installation.
- (c) Reinforcement and concrete.
- (d) Pile records.
- (e) Pile cropping.
- (f) Integrity and load testing.
- (g) Remedial works.
- (h) Completion records.

4.3.4 Below-ground drainage

- (a) Foul drainage.
- (b) Surface-water drainage.
- (c) Bedding and surround.
- (d) Manholes and chambers.
- (e) Connections to existing drainage.
- (f) Rainwater and soil-vent connections.
- (g) Testing and CCTV inspection.
- (h) Backfilling and reinstatement.
- (i) Drainage records.

4.3.5 Below-ground service ducts

- (a) Utility ducts.
- (b) Draw pits.
- (c) Inspection chambers.
- (d) Surface boxes.
- (e) Marker posts and plates.
- (f) Duct fittings and accessories.
- (g) Draw cords and sealing.

- (h) Backfilling and records.

4.3.6 Surface-water attenuation

- (a) Attenuation tank.
- (b) Flow-control device.
- (c) Inlets and outlets.
- (d) Access and maintenance.
- (e) Structural and flotation requirements.
- (f) Backfill and protection.
- (g) Testing and commissioning.

4.3.7 Lift pits

- (a) Excavation.
- (b) Structural base and walls.
- (c) Waterproofing.
- (d) Sumps and drainage.
- (e) Cast-in items.
- (f) Lift interfaces.
- (g) Testing and inspection.

4.3.8 Pile caps and ground beams

- (a) Blinding.
- (b) Formwork.
- (c) Reinforcement.
- (d) Cast-in items and holding-down bolts.
- (e) Concrete and testing.
- (f) Curing.
- (g) Waterproofing interfaces.
- (h) Backfilling.
- (i) Structural records.

4.3.9 Oversite concrete

4.3.10 Below-ground external perimeter walls

- (a) Cavity Wall construction – outer leaf brick, inner leaf block and telescopic vents with M6 Mortar
- (b) Gen 1 mix cavity infill
- (c) interfaces.
- (d) Insulation and protection.
- (e) Drainage.
- (f) Backfilling.
- (g) Sealing of service penetrations.

4.3.11 Below-ground internal Sleeper walls

- (a) Setting out from the approved structural and architectural information.
- (b) Foundations and bearings for sleeper walls where required.
- (c) Blockwork or masonry sleeper walls and support walls.
- (d) Ties, restraints, DPCs and junctions with ground beams and external walls.
- (e) Service openings, sleeves and making good.
- (f) Backfilling and layered compaction following completion of ground beams and sleeper walls.
- (g) Hardcore or granular fill, blinding and formation preparation.
- (h) Ground-gas, radon or vapour-control membranes where required by the approved design.
- (i) Insulation and damp-proof membrane interfaces.
- (j) Oversite concrete, levels, falls, curing and surface tolerances.
- (k) Testing, inspection and record information.

4.3.12 Ground-floor construction

- (a) Approved structural floor system.
- (b) Structural units or slab construction.
- (c) Bearings, reinforcement and infills.
- (d) Concrete topping where required.
- (e) 1200g Damp-proof membrane
- (f) 150mm Insulation and 25mm perimeter insulation
- (g) 500g VCL Layer
- (h) 75mm liquid Screed.
- (i) Movement joints and sealants.
- (j) Service penetrations.
- (k) Threshold interfaces.

4.4 Superstructure

4.4.1 Structural-steel frame

- (a) Columns and beams.
- (b) Bracing and stability systems.
- (c) Connections and base plates.
- (d) Holding-down bolts.
- (e) Secondary steelwork.
- (f) Lift-shaft framing.
- (g) Balcony interfaces.
- (h) Service supports.
- (i) Fire-protection interfaces.
- (j) Fabrication and erection.
- (k) Temporary stability.
- (l) Surveys and records.

4.4.2 Structural-steel fire protection

- (a) Approved intumescent coating, board encasement or other fire-protection system.
- (b) Substrate preparation, primers and compatibility checks.
- (c) Protection to columns, beams, connections, base plates and secondary steelwork.
- (d) Coordination at façades, partitions, ceilings, service penetrations and movement joints.
- (e) Dry-film-thickness checks or system-specific inspection requirements.
- (f) Protection during follow-on works and repair of damage.
- (g) Inspection records, certification and as-built schedule.

4.4.3 Suspended structural floors

- (a) Structural deck or units in accordance with the approved structural design.
- (b) Reinforcement.
- (c) Concrete topping.
- (d) Shear connections.
- (e) Edge trims and restraints.
- (f) Grouting and structural ties.
- (g) Perimeter shuttering.
- (h) In-situ infills.
- (i) Temporary propping and safety netting.
- (j) Openings and penetrations.
- (k) Surface finish.
- (l) Balcony interfaces.

4.4.4 Precast stairs

- (a) Stair flights and landings.

- (b) Bearings.
- (c) Lifting and installation.
- (d) Temporary guarding.
- (e) Grouting and connections.
- (f) Finishes and balustrade interfaces.

4.4.5 Structural robustness and movement

- (a) Horizontal and vertical ties.
- (b) Load paths.
- (c) Movement joints.
- (d) Deflection allowances.
- (e) Façade movement interfaces.
- (f) Partition and fire-stopping interfaces.

4.4.6 Balcony structural interfaces

- (a) Structural connections.
- (b) Thermal breaks.
- (c) Waterproofing.
- (d) Drainage.
- (e) Balustrade fixings.
- (f) Decking supports.
- (g) Fire-performance requirements.

4.4.7 Service and riser openings

- (a) Structural openings.
- (b) Trimmer steelwork.
- (c) Cast-in sleeves.
- (d) Penetration setting out.
- (e) M&E coordination.
- (f) Fire and acoustic sealing.
- (g) Maintenance access.

4.5 Building Envelope

4.5.1 External-wall build-ups

- (a) Internal linings.
- (b) DuPont AirGuard Control - AVCL. (Fire Rated)
- (c) SFS framing
- (d) Insulation
- (e) Siniat weather defence board.
- (f) Duoslab Insulation.
- (g) Stainless steel channel and ties
- (h) Floor, roof, window, door and balcony interfaces
- (i) Airtightness sealants EPDM
- (j) Air- and weather-tightness.

4.5.2 Masonry

- (a) Facing brickwork.
- (b) Mortar.
- (c) Wall ties and joint reinforcement.
- (d) Supports, shelf angles and lintels.
- (e) Sills and parapet copings.
- (f) Movement joints.
- (g) Fire-sealed head joints.

- (h) Sealants, cleaning and protection.

4.5.3 Cavity barriers and fire stopping

- (a) (a) Horizontal and vertical cavity barriers.
- (b) (b) Fire barriers around openings and compartment lines.
- (c) (c) Fire stopping at floor edges and façade junctions.
- (d) (d) Fire-barrier inspections and records.

4.5.4 DPCs, cavity trays and weep vents

- (a) Horizontal DPCs.
- (b) Horizontal cavity trays.
- (c) Stop ends.
- (d) Laps and joints.
- (e) Weep vents.
- (f) DPCs around openings.
- (g) SFS, masonry and waterproofing interfaces.
- (h) External-level and threshold coordination.
- (i) Inspection before concealment.

4.5.5 Masonry movement joints

- (a) Horizontal head joints.
- (b) Vertical movement joints.
- (c) Joint filler.
- (d) Fire-resistant seals.
- (e) External weather sealant.
- (f) Structural movement coordination.
- (g) Façade and fire-stopping interfaces.

4.5.6 Rainscreen cladding

- (a) Carrier frame and brackets.
- (b) Support rails and fixings.
- (c) Thermal isolators.
- (d) Insulation.
- (e) Cavity barriers.
- (f) Breather membrane.
- (g) Cladding panels.
- (h) Flashings and closures.
- (i) Window, door, roof and masonry interfaces.
- (j) Fire and wind-load requirements.

4.5.7 Windows and external doors

- (a) Windows and external doors.
- (b) Balcony and roof-access doors.
- (c) Fire-exit doors.
- (d) Frames, glazing and ironmongery.
- (e) Safety and restrictor devices.
- (f) AOV interfaces.
- (g) Cills, jambs, heads and perimeter closures.
- (h) Trickle ventilators and restrictors.
- (i) Perimeter insulation, air seals and weather seals.
- (j) Ironmongery and safety devices.
- (k) Cavity closers and trays.
- (l) Air- and weather-sealing.
- (m) Thresholds and making good.

- (n) Testing.

4.5.8 Curtain walling

- (a) Framing and glazing.
- (b) Doors and panels.
- (c) Brackets and fixings.
- (d) Perimeter fire stopping.
- (e) Air- and weather-sealing.
- (f) Drainage.
- (g) Structural and façade interfaces.
- (h) Testing and certification.

4.5.9 Balconies

- (a) Structural interfaces.
- (b) Decking and supports.
- (c) Falls and drainage.
- (d) Waterproofing.
- (e) Balustrades.
- (f) Fire-performance requirements.
- (g) Threshold coordination.
- (h) Finishes and maintenance access.

4.5.10 Roofing

- (a) Roof deck and substrate.
- (b) Insulation.
- (c) Vapour-control layer.
- (d) Waterproof membrane.
- (e) Upstands.
- (f) Outlets, gutters and downpipes.
- (g) Fascias.
- (h) Parapets and copings.
- (i) Roof-access doors and hatches.
- (j) Plant bases and penetrations.
- (k) Walkways and fall-arrest systems.
- (l) Lightning-protection interfaces.
- (m) Inspection and testing.

4.5.11 Façade and plantroom louvres

- (a) Intake and extract louvres, frames, blades, supports and fixings.
- (b) Weathering, drainage, perimeter sealing, bird and insect mesh and acoustic treatment where required.
- (c) Fire-performance, cavity-barrier, façade, ductwork and plant interfaces.
- (d) Internal closures, finishes and access for inspection, cleaning and replacement.
- (e) Testing, certification and record information.

4.6 Building Services

4.6.1 Incoming utilities

- (a) Incoming electricity.
- (b) Incoming water.
- (c) Telecommunications and FTTP.
- (d) Public-sewer connections.
- (e) Utility ducts and chambers.
- (f) Metering.
- (g) Authority interfaces.

- (h) Testing and connection records.
- (i) Identification, isolation, protection, diversion and reconnection of existing utilities.
- (j) Separate domestic and sprinkler incoming water services, including provider-approved barrier pipework.
- (k) External service trenches, draw pits, markers, backfilling and reinstatement.
- (l) ENWL non-contestable and ICP contestable work boundaries.
- (m) Domestic incoming water service using 90 mm barrier pipework designed for 6 l/s, subject to final United Utilities approval.
- (n) Sprinkler incoming water service using 63 mm barrier pipework designed for 3 l/s, subject to final United Utilities approval.
- (o) Point-of-connection works, statutory-provider inspections, permanent connections and energisation.
- (p) Utility connection, adoption, metering and completion records.

4.6.2 DNO substation

- (a) Building shell and foundations.
- (b) Walls and roof.
- (c) Secure doors.
- (d) Ventilation and finishes.
- (e) DNO equipment interfaces.
- (f) Earthing.
- (g) Earthing study, DNO/ICP submission and approval.
- (h) Incoming and outgoing ducts.
- (i) Access and maintenance.

4.6.3 Electrical distribution

- (a) Intake equipment and main switchgear.
- (b) Landlord and apartment distribution.
- (c) Risers and containment.
- (d) Cabling.
- (e) Metering.
- (f) Earthing and bonding.
- (g) Labels and schedules.

4.6.4 Water services

- (a) Incoming water main.
- (b) Water-storage tank.
- (c) Water treatment.
- (d) Booster pumps.
- (e) Landlord and apartment distribution.
- (f) Valves and insulation.
- (g) Water meters.
- (h) Testing, flushing, cleaning, chlorination and disinfection, including sampling, results, certification and records in accordance with the approved water-hygiene procedure and statutory-provider requirements.

4.6.5 Fire alarm and life-safety cabling

- (a) Fire-alarm panels.
- (b) Detection.
- (c) Sounders and visual alarms.
- (d) Apartment and common-area interfaces.
- (e) Interfaces with smoke control, sprinklers, lifts and access control.
- (f) Life-safety cabling.
- (g) Cause-and-effect programming.
- (h) Testing and certification.

4.6.6 Smoke control and AOV systems

- (a) Stair AOVs.
- (b) Corridor smoke-clearance systems.
- (c) Smoke shafts.
- (d) Fans and dampers.
- (e) Control panels.
- (f) Fire-alarm interfaces.
- (g) Standby-power interfaces.
- (h) Make-up air.
- (i) Builders' work and openings.
- (j) Cause-and-effect testing.
- (k) Commissioning and certification.

4.6.7 Dry-riser system

- (a) Inlet breeching.
- (b) Riser pipework.
- (c) Landing valves and cabinets.
- (d) Drainage.
- (e) Signage.
- (f) Fire stopping.
- (g) Testing and certification.

4.6.8 Sprinkler system

- (a) Incoming supply.
- (b) Storage and pumps where required.
- (c) Apartment and common-area distribution.
- (d) Monitored floor valves.
- (e) Flow switches.
- (f) Fire-alarm interfaces.
- (g) Drain and test points.
- (h) Testing and certification.

4.6.9 Solar PV and SolShare

- (a) PV modules and mounting frames.
- (b) Roof fixings.
- (c) Inverters.
- (d) DC and AC cabling.
- (e) Isolators.
- (f) SolShare equipment.
- (g) Main-board connection.
- (h) Monitoring and export limitation.
- (i) G99/G100 requirements.
- (j) Testing and commissioning.

4.6.10 Electric-vehicle charging

- (a) Charging units and mounting arrangements.
- (b) Dedicated distribution, protective devices, cabling and containment.
- (c) Below-ground ducts, draw pits, bases and builders' work.
- (d) Load management and diversity controls.
- (e) Metering, communications, access and revenue-control arrangements.
- (f) Earthing, impact protection, signage and bay identification.
- (g) Testing, commissioning, user information and warranties.

4.6.11 Passenger lifts

- (a) Lift cars and machinery.
- (b) Controls.
- (c) Landing doors.
- (d) Indicators and controls.
- (e) Fire and emergency interfaces.
- (f) Power and standby-power interfaces.
- (g) Firefighting-lift or evacuation-lift power requirements where required by the approved fire strategy.
- (h) Shaft equipment.
- (i) Builders' work.
- (j) Finishes.
- (k) Testing and certification.

4.6.12 Communications systems

- (a) Telephone and data.
- (b) TV and satellite.
- (c) Roof aerial and satellite equipment.
- (d) Containment.
- (e) Apartment and landlord interfaces.
- (f) Testing.

4.6.13 Security systems

- (a) Internal and external CCTV.
- (b) Access control.
- (c) Door-entry systems.
- (d) Fobs and access credentials.
- (e) Power supplies and containment.
- (f) Testing and commissioning.

4.6.14 Lightning and surge protection

- (a) Air-termination network.
- (b) Down conductors.
- (c) Earth termination.
- (d) Surge-protection devices.
- (e) Roof and façade interfaces.
- (f) Testing and certification.

4.6.15 Standby generator and flue

- (a) Confirmed life-safety load schedule.
- (b) Generator set.
- (c) Automatic transfer switch.
- (d) Life-safety distribution.
- (e) Fire-rated cabling.
- (f) Fuel system and storage where required.
- (g) Acoustic enclosure.
- (h) Vibration isolation.
- (i) Cooling and ventilation.
- (j) Exhaust flue, supports and weathering.
- (k) Fire compartmentation.
- (l) Bases and plinths.
- (m) Earthing.
- (n) Controls and alarms.
- (o) Fire-alarm, smoke-control, lift and sprinkler interfaces.
- (p) Load-bank and cause-and-effect testing.

- (q) Commissioning and certification.

4.6.16 Above-ground drainage

- (a) Soil vent pipes and waste pipes.
- (b) Branch connections.
- (c) Roof-drainage interfaces.
- (d) Access and rodding points.
- (e) Acoustic treatment.
- (f) Fire stopping.
- (g) Below-ground connections.
- (h) Testing.

4.6.17 Ventilation

- (a) Apartment ventilation in accordance with the approved design.
- (b) Kitchen, bathroom and WC extract dMEV
- (c) Background ventilation and trickle ventilators where required by the approved Part F strategy.
- (d) Fresh-air provision independent of any recirculating heating or cooling system.
- (e) Ductwork and terminals.
- (f) Fire and smoke dampers.
- (g) Acoustic treatment.
- (h) Condensate drainage.
- (i) Controls and power.
- (j) Cleaning and maintenance access.
- (k) Testing and commissioning.

4.6.18 Power and lighting

- (a) Socket outlets and fused connections.
- (b) Internal lighting.
- (c) Feature lighting.
- (d) Emergency lighting.
- (e) External lighting.
- (f) Appliance and mechanical-equipment supplies.
- (g) Final connections.
- (h) Testing and certification.

4.6.19 Apartment space heating

- (a) Approved apartment space-heating system.
- (b) Electric panel heaters in accordance with the approved strategy.
- (c) Refrigerant pipework, insulation, containment and pressure testing where applicable.
- (d) Condensate collection and drainage.
- (e) Room controls, thermostats, isolators and electrical supplies.
- (f) Façade penetrations, fire and acoustic sealing and weatherproofing.
- (g) Noise, overheating, SAP/EPC and maintenance-access coordination.
- (h) Testing, commissioning, resident guidance and warranties.

4.6.20 Domestic hot and cold water

- (a) Hot-water cylinders.
- (b) Hot- and cold-water distribution.
- (c) Valves and insulation.
- (d) Sanitaryware and kitchen connections.
- (e) Temperature control.
- (f) Testing and commissioning.

4.6.21 Wastewater heat recovery where required

- (a) Wastewater heat-recovery units serving baths or other approved appliances where required by the approved SAP and energy strategy.
- (b) Waste, drainage and cold-water connections.
- (c) Supports, access panels, maintenance access, acoustic treatment, insulation and condensation control.
- (d) Coordination with floor construction, bath supports and drainage falls.
- (e) Testing, commissioning, certification, warranties and O&M information.

4.6.22 Permanent fire extinguishers

- (a) Fire-risk-based extinguisher schedule.
- (b) Extinguisher types and capacities.
- (c) Common-area and plantroom extinguishers.
- (d) Wall brackets or cabinets.
- (e) Identification signs.
- (f) Inspection and certification.
- (g) Asset labels.
- (h) Maintenance information.

4.6.23 Builders' work in connection

- (a) Setting out holes, mortices and chases.
- (b) Pipe and duct sleeves.
- (c) Bases and plinths.
- (d) Secondary supports.
- (e) Catenary cables.
- (f) Cutting and making good where required.
- (g) External service fittings and accessories.
- (h) Inspection chambers.
- (i) Surface boxes and service pits.
- (j) Marker posts and plates.
- (k) Final service connections.
- (l) Fire, acoustic and weather sealing.
- (m) Coordination and record information.

4.7 Internal Construction and Fit-Out

4.7.1 Masonry partitions

- (a) Blockwork partitions.
- (b) Ties and restraints.
- (c) Lintels.
- (d) Movement joints.
- (e) Fire stopping.
- (f) Acoustic sealing.
- (g) Openings and making good.

4.7.2 Primary metal framing

- (a) Load-bearing and non-load-bearing framing.
- (b) Corridor and party walls.
- (c) Shaft walls.
- (d) Head and base tracks.
- (e) Deflection heads.
- (f) Reinforcement and pattressing.
- (g) Fire and acoustic insulation.
- (h) Door and service coordination.

4.7.3 Screeds and acoustic layers

- (a) Acoustic resilient layers.
- (b) Perimeter isolation.
- (c) Thermal insulation.
- (d) Screeds and reinforcement.
- (e) Movement joints and falls.
- (f) Moisture testing.
- (g) Surface preparation.

4.7.4 Pattressing and access panels

- (a) Pattressing for sanitaryware, kitchens, handrails, FF&E and M&E equipment.
- (b) Pattressing for access control and signage.
- (c) Fire-rated access panels.
- (d) Acoustic access panels.
- (e) Moisture-resistant and tiled access panels.
- (f) Identification and schedules.

4.7.5 Drylining

- (a) Wall linings and partitions.
- (b) Shaft-wall boarding.
- (c) Ceiling boarding.
- (d) Fire-rated boards.
- (e) Acoustic boards.
- (f) Moisture-resistant boards.
- (g) Insulation.
- (h) Jointing and taping.
- (i) Service penetrations and making good.

4.7.6 Ceiling systems

- (a) MF or other approved ceiling framing, hangers and supports.
- (b) Plasterboard, suspended-tile or specialist ceiling finishes in accordance with the approved design.
- (c) Fire, acoustic and moisture-resistance requirements.
- (d) Bulkheads, margins, shadow gaps and perimeter details.
- (e) Access panels and removable sections.
- (f) Coordination with lighting, sprinklers, detectors, ventilation and other ceiling services.
- (g) Fire stopping, acoustic sealing, decoration and completion records.

4.7.7 Pipe and service boxing

- (a) Boxing to soil, waste, water and ventilation services.
- (b) Framing and plasterboard.
- (c) Moisture-resistant boards.
- (d) Fire-rated boards.
- (e) Acoustic insulation.
- (f) Access panels.
- (g) Valve and rodding access.
- (h) Fire stopping.
- (i) Decoration-ready finish.
- (j) Kitchen, bathroom and M&E coordination.

4.7.8 Bathroom tanking

- (a) Substrate preparation.
- (b) Tanking membranes.
- (c) Primers and reinforcing tapes.
- (d) Corners and penetrations.

- (e) Bath and shower interfaces.
- (f) Testing and protection.

4.7.9 Fire stopping and compartmentation

- (a) Linear-gap seals.
- (b) Penetration seals.
- (c) Cavity barriers.
- (d) Fire collars and wraps.
- (e) Fire-rated compounds.
- (f) Movement seals.
- (g) Fire-rated access panels.
- (h) Identification and inspections.
- (i) Photographic and digital evidence.
- (j) Regulation 38 information.

4.7.10 Plastering

- (a) Wall and ceiling skim finishes.
- (b) Beads, stops and movement joints.
- (c) Joint, corner and substrate preparation.
- (d) Reveals, openings and service penetrations.
- (e) Making good and patch repairs.
- (f) Protection of completed plasterwork.

4.7.11 Tiling

- (a) Wall and floor tiling.
- (b) Substrate preparation and levelling.
- (c) Tanking and waterproofing interfaces.
- (d) Adhesives, grout and primers.
- (e) Tile trims, movement joints and sealants.
- (f) Setting out, cuts and service interfaces.
- (g) Cleaning and protection of completed tiling.

4.7.12 Decoration

- (a) Surface preparation and filling.
- (b) Primers, sealers and undercoats.
- (c) Paint finishes to walls and ceilings.
- (d) Paint finishes to woodwork and metalwork.
- (e) Decoration to apartments, communal corridors, lobbies, reception and circulation areas.
- (f) Final making good and touch-up decoration.
- (g) Protection and final cleaning.

4.7.13 Internal doors and ironmongery

- (a) Apartment entrance doors.
- (b) Internal apartment doors.
- (c) Communal and service-riser doors.
- (d) Fire doors.
- (e) Frames, linings and architraves.
- (f) Ironmongery.
- (g) Access-control interfaces.
- (h) Fire and acoustic seals.
- (i) Door signage.
- (j) Certification.

4.7.14 General joinery

- (a) Skirtings and architraves.
- (b) Window boards and trims.
- (c) Utility cupboards and storage-unit joinery shown on the approved drawings.
- (d) Sundry joinery.
- (e) Finishes and fixings.

4.7.15 Kitchens

- (a) Kitchen units and worktops.
- (b) Splashbacks.
- (c) Sinks, taps and wastes.
- (d) Hobs, ovens and extractors.
- (e) Appliances.
- (f) Plumbing and electrical interfaces.
- (g) Sealants, testing and protection.

4.7.16 Bathrooms and sanitaryware

- (a) Baths and showers.
- (b) WCs and washbasins.
- (c) Taps and wastes.
- (d) Shower screens and accessories.
- (e) Supports and pattressing.
- (f) Plumbing connections.
- (g) Sealants and testing.
- (h) Accessibility requirements.

4.7.17 Floor finishes

- (a) Substrate preparation.
- (b) Moisture testing.
- (c) Levelling compounds.
- (d) LVT or laminate.
- (e) Carpet.
- (f) Vinyl.
- (g) Threshold strips.
- (h) Protection and final cleaning.

4.7.18 FF&E

- (a) Apartment mailboxes.
- (b) Mailbox numbering.
- (c) Reception desk.
- (d) Estate-management desk.
- (e) Noticeboards.
- (f) Statutory and sundry signage.
- (g) Communal fittings.
- (h) Fixings and pattressing.
- (i) Power and data interfaces.
- (j) Accessibility and fire-performance requirements.

4.7.19 Building signage and wayfinding

- (a) Apartment numbering.
- (b) Floor and core identification.
- (c) Statutory and fire-safety signage.
- (d) Directional and accessible signage.
- (e) Plantroom and service-room signage.

- (f) External wayfinding.
- (g) Fixings and schedules.

4.7.20 Temporary protection of completed works

- (a) Protection to floors, doors, windows, façades, sanitaryware, kitchens, lifts and completed finishes.
- (b) Inspection, maintenance and replacement of damaged protection.
- (c) Removal before handover and repair of damage caused by follow-on works.

4.7.21 Reception, parcel store, communal WC and communal spaces

- (a) Complete reception-area construction and fit-out.
- (b) Parcel-store walls, doors, finishes, shelving, storage racks or parcel lockers where scheduled.
- (c) Communal-WC sanitaryware, plumbing, drainage, ventilation, lighting and finishes.
- (d) Communal spaces at each applicable floor.
- (e) Floors, walls, ceilings, decoration, joinery, impact protection and corner protection.
- (f) Doors, ironmongery, access control, mailboxes, reception desk, noticeboards and signage.
- (g) Lighting, emergency lighting, power, data, CCTV, fire alarm and sprinkler coverage where required.
- (h) Accessible entrances, thresholds and circulation space.
- (i) Fire stopping, compartmentation, testing, final cleaning, protection and handover.

4.8 External Works

4.8.1 Surface-water drainage

- (a) External drainage pipes.
- (b) Channels and gullies.
- (c) Catchpits and manholes.
- (d) Attenuation interfaces.
- (e) Outfalls.
- (f) Testing and CCTV inspection.
- (g) Reinstatement.

4.8.2 Retaining walls

- (a) Setting out.
- (b) Excavation and disposal.
- (c) Hardcore and blinding.
- (d) Structural foundations.
- (e) Reinforced-concrete or approved retaining-wall construction.
- (f) Formwork and reinforcement.
- (g) Waterproofing or tanking.
- (h) Land drainage, filter media and weep holes.
- (i) Granular backfill and layered compaction.
- (j) Copings and finishes.
- (k) Handrails and guarding where required.
- (l) Temporary works and excavation support.
- (m) Canal-side and ground-condition interfaces.
- (n) Car-parking, kerb, paving and drainage interfaces.
- (o) Garden, boundary and landscaping interfaces.
- (p) Testing, inspection and completion records.

4.8.3 External foul drainage and washdown drainage

- (a) External foul drainage branches and connections.
- (b) Trapped foul floor gullies or drains to each external bin store.
- (c) Bin-store floor falls, channels where required and washdown drainage.
- (d) Water points or hose connections where required by the approved cleaning strategy.
- (e) Rodding and maintenance access.

- (f) Testing, CCTV inspection and record drawings.

4.8.4 Roads and parking

- (a) Excavation and sub-base.
- (b) Kerbs.
- (c) Asphalt construction.
- (d) Permeable paving or permeable pavement construction where required by the approved drainage design.
- (e) Flush kerbs, dropped kerbs and transitions.
- (f) Private-driveway construction and interfaces where included within the current contract boundary.
- (g) Parking bays and accessible parking.
- (h) Line marking.
- (i) Vehicle barriers and bollards.
- (j) Drainage interfaces.
- (k) Signage and making good.

4.8.5 Paving and footpaths

- (a) Excavation and sub-base.
- (b) Flag and stone paving.
- (c) Kerbs and edgings.
- (d) Bullnose kerbs and special kerb profiles shown on the approved external-works drawings.
- (e) Secure-by-design symbolic barrier formed as a flush block-paved strip where required.
- (f) Tactile paving.
- (g) Falls and drainage.
- (h) Accessible routes.
- (i) Making good.

4.8.6 Section 278, highway and third-party works

- (a) Highway-authority approvals, licences, permits and inspections where required.
- (b) Tie-ins to adopted carriageways, footways, crossings, kerbs and drainage.
- (c) Traffic management, temporary works and protection of the public.
- (d) Third-party land access, licences and reinstatement.
- (e) Completion certificates and adoption records where applicable.

4.8.7 Cycle storage

- (a) Foundations and enclosure.
- (b) Structure and roof.
- (c) Doors.
- (d) Cycle racks.
- (e) Lighting and access control where required.
- (f) Signage and drainage.

4.8.8 Bin stores

- (a) Foundations.
- (b) Walls and roofs.
- (c) Louvered doors.
- (d) Drainage and ventilation.
- (e) Finishes and signage.
- (f) Access and cleaning requirements.
- (g) Washable floor and wall finishes.
- (h) Trapped foul floor drainage and graded falls.
- (i) Washdown water supply where required.
- (j) Rodding access, cleaning access and pest-resistant detailing.

4.8.9 External signage and furniture

- (a) Reflective traffic signs.
- (b) Illuminated traffic signs.
- (c) Signposts and foundations.
- (d) Electrical supplies.
- (e) Benches.
- (f) Pillar boxes or external service enclosures as defined by the approved design.
- (g) Fixings.
- (h) Utility, drainage and landscaping coordination.
- (i) External wayfinding.

4.8.10 External lighting infrastructure

- (a) Lighting columns, bollards or wall-mounted fittings in accordance with the approved lighting design.
- (b) Column foundations, root protection and impact protection.
- (c) Below-ground ducts, draw pits, cabling routes and earthing.
- (d) Controls, photocells, time clocks and connection to the landlord supply.
- (e) Coordination with roads, parking, drainage, utilities and landscaping.
- (f) Testing, commissioning and record information.

4.8.11 Fencing and boundary treatments

- (a) Chain-link fencing.
- (b) Metal post-and-rail fencing.
- (c) Timber fencing.
- (d) Railings and gates.
- (e) Boundary walls and copings.
- (f) Foundations and finishes.
- (g) Retaining-wall and landscape interfaces.
- (h) 2100 mm-high metal railings where shown on the current approved boundary plan.
- (i) 2100 mm-high wall-and-metal-railing boundary construction where shown.
- (j) 600 mm-high retaining wall or boundary structure where shown.
- (k) Retaining walls with metal railings not less than 1500 mm high where shown.
- (l) 2100 mm-high and 1500 mm-high timber fencing where shown.
- (m) 1500 mm-high, 1200 mm-high and 1100 mm-high metal railings where shown.
- (n) Retention, protection, repair and making good to existing boundary walls over 2 m high.
- (o) Matching secure gates, locks, closers and access-control interfaces.
- (p) Secure-by-design and Crime Impact Statement requirements.

4.8.12 Soft landscaping

- (a) Topsoil.
- (b) Turfing and seeding.
- (c) Shrub and tree planting.
- (d) Defensive planting.
- (e) Planning-required planting.
- (f) Private-garden topsoil, turfing, planting, paving interfaces and access.
- (g) Landscaping outside the site boundary where required by planning or third-party agreement.
- (h) Tree pits, root barriers, structural soils and service-clearance requirements.
- (i) Tree stakes, ties, guards and below-ground anchoring.
- (j) Soil improvement, fertiliser, mulch and weed control.
- (k) Watering and irrigation provisions required for establishment.
- (l) Defensive-planting scope, planting schedules and Crime Impact Statement compliance.
- (m) Protection and making good to completed house plots, private drives and adjoining landscape areas.
- (n) Establishment maintenance.
- (o) Replacement of failed planting.

(p) Completion records.

5.0 TESTING, COMMISSIONING AND CERTIFICATION

5.1 Mechanical Testing and Commissioning

5.1.1 Water systems

- (a) Flushing, cleaning, chlorination and disinfection, including sampling, laboratory results and certification.
- (b) Pressure testing.
- (c) Water-hygiene and legionella compliance.
- (d) Water Risk Assessment and water-hygiene management information.
- (e) Booster and storage commissioning.
- (f) Meter verification.
- (g) Domestic hot-water commissioning.

5.1.2 Heating and cooling systems

- (a) Functional testing of emitters, indoor units and external units.
- (b) Controls, thermostat and electrical-interlock testing.
- (c) Refrigerant pressure, leak and F-Gas records where applicable.
- (d) Condensate drainage testing.
- (e) Noise and vibration checks.
- (f) Commissioning records and resident demonstrations.

5.1.3 Drainage systems

- (a) Above- and below-ground drainage testing.
- (b) CCTV drainage surveys.
- (c) Confirmation of clear operation.
- (d) Drainage record drawings.

5.1.4 Ventilation

- (a) Ductwork leakage testing.
- (b) Airflow testing and balancing.
- (c) Acoustic testing.
- (d) Controls testing.
- (e) Part F compliance.
- (f) Commissioning records.

5.1.5 Sprinklers and dry risers

- (a) Pressure and flow testing.
- (b) Alarm-interface testing.
- (c) Sprinkler certification.
- (d) Dry-riser pressure testing and certification.

5.1.6 Wastewater heat recovery where required

- (a) Functional, flow and leakage testing of installed wastewater heat-recovery units.
- (b) Verification of drainage and cold-water connections and maintenance access.
- (c) Commissioning records and product certification.

5.2 Electrical Testing and Commissioning

5.2.1 Electrical installations

- (a) Dead and live testing.
- (b) Distribution-board testing.
- (c) Earthing and bonding verification.
- (d) Circuit schedules.
- (e) Electrical-installation certification.

5.2.2 Lighting

- (a) Functional testing.
- (b) Lighting-control testing.
- (c) Emergency-lighting duration testing.
- (d) External-lighting testing.
- (e) Certification.

5.2.3 Renewable energy

- (a) PV, inverter and SolShare commissioning.
- (b) Monitoring-system commissioning.

5.2.4 Electric-vehicle charging

- (a) Charger functional and load-management testing.
- (b) Electrical certification, communications, metering and payment-system testing.
- (c) Asset records, warranties and user information.
- (d) G99/G100 certification.
- (e) Export-control testing.

5.2.5 Security and communications

- (a) CCTV testing.
- (b) Access-control and intercom testing.
- (c) Data and telecommunications testing.
- (d) TV and satellite testing.
- (e) User demonstrations.

5.2.6 Standby generator

- (a) Generator start-up.
- (b) Automatic changeover testing.
- (c) Load-bank testing.
- (d) Life-safety load testing.
- (e) Alarm and control testing.
- (f) Fuel, flue and ventilation checks.
- (g) Cause-and-effect testing.
- (h) Certification.

5.3 Fire and Life-Safety Testing

5.3.1 Fire alarm

- (a) Device and sound-level testing.
- (b) Cause-and-effect testing.
- (c) Interfaces with lifts, access control, sprinklers, ventilation and smoke control.
- (d) Certification.

5.3.2 Passive fire protection

- (a) Fire-stopping inspections.
- (b) Compartmentation inspections.
- (c) Cavity-barrier inspections.
- (d) Fire-door inspections.
- (e) Evidence review.
- (f) Structural-steel fire-protection inspection and certification.
- (g) Regulation 38 sign-off.

5.3.3 Smoke control and AOV

- (a) Smoke-control and AOV testing.
- (b) Fan and damper testing.

- (c) Standby-power testing.
- (d) Cause-and-effect testing.
- (e) Integrated fire-strategy testing.
- (f) Certification.

5.3.4 Permanent fire extinguishers

- (a) Installation inspection.
- (b) Equipment certification.
- (c) Asset labelling.
- (d) Maintenance requirements.

5.4 General and Statutory Compliance

5.4.1 Building performance

- (a) Air-tightness testing.
- (b) Acoustic testing.
- (c) Thermal-performance evidence.
- (d) Final SAP calculations.
- (e) EPCs.
- (f) Part F, Part G and Part O evidence as applicable.
- (g) External-works levels, falls, drainage, boundary, lighting and landscape inspections.

5.4.2 Lifts

- (a) Lift commissioning.
- (b) Safety-system testing.
- (c) Emergency-operation testing.
- (d) LOLER certification.
- (e) Facilities-management training.

5.4.3 Utilities and authorities

- (a) Utility-authority inspections.
- (b) DNO approval.
- (c) Water and sewer approvals.
- (d) Telecommunications completion.
- (e) Metering records.
- (f) Building Control inspections and approvals.

5.4.4 Witness testing and records

- (a) Witness testing of critical systems.
- (b) Test sheets and commissioning records.
- (c) Certification register.
- (d) Defect and retest records.
- (e) Final commissioning report.

5.4.5 Accessibility and project-specific completion verification

- (a) Verify the completed dwelling-accessibility schedule and inspect M4(1), M4(2) and M4(3) apartments where applicable.
- (b) Verify accessible entrances, thresholds, circulation routes, controls and parking spaces.
- (c) Inspect the reception, parcel store, communal WC and floor-by-floor communal spaces.
- (d) Verify completion against the approved Schedule of Accommodation and record defects, remedial works and final compliance evidence.

6.0 HANDOVER AND CLOSE-OUT

6.1 Practical Completion

6.1.1 Snagging and defect rectification

- (a) Contractor, consultant and Employer's Agent inspections.
- (b) Snagging records.
- (c) Defect rectification.
- (d) Retesting.
- (e) Closure evidence.

6.1.2 Builders' clean, façade clean and final cleaning

- (a) Apartments and communal areas.
- (b) Plantrooms.
- (c) Windows and façade.
- (d) External areas.
- (e) Removal of labels, protection and waste.
- (f) Progressive and final builders' clean to apartments, communal areas, reception, parcel store, service rooms and plantrooms.
- (g) Final cleaning of floors, walls, ceilings, doors, sanitaryware, kitchens, appliances, fittings and equipment.
- (h) Internal and external cleaning of windows, glazing, curtain walling, cladding, brickwork and other façade surfaces.
- (i) Specialist access equipment required for the façade clean.
- (j) Removal of mortar, sealant, adhesive, labels, protective films and construction residue without damaging completed surfaces.
- (k) Cleaning of external roads, paving, drainage channels, bin stores, cycle stores and landscaped interfaces.

6.1.3 Practical Completion

- (a) Pre-completion inspection.
- (b) Completion of statutory requirements.
- (c) Completion of life-safety systems.
- (d) Confirmation that building systems are operational.
- (e) Practical Completion inspection and certification.
- (f) Record of outstanding minor works.
- (g) Evidence of registered postal addresses for each apartment.
- (h) Final schedule of accommodation and unit references.
- (i) Tenancy or ownership information provided by the Employer where required for handover.
- (j) Final dated site-photograph record covering the building, external works and completed interfaces.

6.2 Statutory Approvals and Warranties

6.2.1 Planning

- (a) Planning-condition discharge.
- (b) Pre-occupation condition discharge.
- (c) Planning satisfaction letters.
- (d) Approved planning-document register.

6.2.2 Building Control

- (a) Final inspections.
- (b) Dutyholder declarations.
- (c) Completion information.
- (d) Building Control Final Certificate.
- (e) Cavity-wall approval evidence where required.

6.2.3 Structural warranty

- (a) Closure of warranty comments.
- (b) Final inspection.
- (c) Cover note.
- (d) Policy wording.
- (e) Warranty certificates for each dwelling.

6.2.4 Collateral warranties and insurance

- (a) Professional-team warranties.
- (b) Key-subcontractor warranties.
- (c) Reliance letters.
- (d) Professional indemnity evidence.
- (e) Contractor and subcontractor insurance evidence.

6.2.5 Legal agreements and third-party consents

- (a) Highway agreements and completion certificates under Sections 38 and 278 where applicable.
- (b) Sewer connection, adoption and diversion agreements under Sections 104, 106 and 185 where applicable.
- (c) DNO substation lease, access arrangements, utility easements and wayleaves.
- (d) Canal-side authority or landowner approvals.
- (e) Third-party land-access agreements, licences, off-site landscaping approvals and maintenance agreements.
- (f) Utility adoption and connection documentation and register of outstanding legal or third-party obligations.

6.3 O&M, As-Built and Health and Safety Information

6.3.1 Health and Safety File

- (a) Design and residual-risk information.
- (b) As-built drawings.
- (c) Product and material information.
- (d) Maintenance requirements.
- (e) Access and replacement information.
- (f) Statutory certification.

6.3.2 Operation and Maintenance manuals

- (a) Architectural information.
- (b) Structural information.
- (c) Mechanical and electrical information.
- (d) Fire and life-safety information.
- (e) Manufacturer literature.
- (f) Final fit-out specification, finishes schedule and product/colour schedule.
- (g) Maintenance schedules and spares.
- (h) Supplier and installer details.
- (i) Commissioning and training records.

6.3.3 As-built information

- (a) Architectural, structural and civil drawings.
- (b) Drainage drawings.
- (c) Mechanical and electrical drawings.
- (d) Fire-strategy information.
- (e) Fire-stopping records.
- (f) Asset registers and equipment schedules.
- (g) Record models where required.

6.4 Fire and Life-Safety Handover

6.4.1 Fire-safety documents

- (a) Final fire strategy and as-built fire plans.
- (b) Regulation 38 information.
- (c) Fire Risk Assessment.
- (d) FRAEW and EWS1 where required.
- (e) Fire Action Plans.
- (f) Building Safety Manual where applicable.

6.4.2 Fire-system certification

- (a) Fire-alarm certification.
- (b) Smoke-detection certification.
- (c) Sprinkler and dry-riser certification.
- (d) AOV and smoke-control certification.
- (e) Emergency-lighting certification.
- (f) Fire-door certification.
- (g) Passive-fire and fire-stopping evidence.
- (h) Confirmation of apartment and communal fire-alarm connections and interfaces to the main fire-alarm system.
- (i) Structural-steel fire-protection records and certification.
- (j) Permanent fire-extinguisher records.

6.5 System Certification

6.5.1 Electrical records

- (a) NICEIC and electrical-installation certificates.
- (b) Emergency-lighting certification.
- (c) Lightning-protection certification.
- (d) PV and SolShare certification.
- (e) G99/G100 approval.
- (f) Generator commissioning certification.
- (g) Electric-vehicle charging certification and commissioning records.

6.5.2 Mechanical records

- (a) Water-hygiene and legionella records.
- (b) Pressure-test records.
- (c) Flushing, chlorination and disinfection records, including sampling and laboratory results.
- (d) Ventilation commissioning.
- (e) Heating and cooling commissioning records.
- (f) Water Risk Assessment.
- (g) Sprinkler, dry-riser and smoke-control records.
- (h) Plantroom commissioning.
- (i) Wastewater heat-recovery commissioning, certification and maintenance records where installed.

6.5.3 Building-performance records

- (a) Final SAP calculations and EPCs.
- (b) Part F, Part G and Part O evidence.
- (c) Air-tightness reports.
- (d) Acoustic reports.
- (e) Thermal-performance evidence.

6.5.4 Civil, structural and envelope records

- (a) Structural calculations.
- (b) Pile and retaining-wall records.

- (c) Drainage CCTV survey.
- (d) Drainage as-built information.
- (e) Roof inspection records and warranties.
- (f) Fall-arrest certification.
- (g) Façade and window testing.
- (h) Structural guarantees.

6.6 Operational Handover

6.6.1 Utility information

- (a) Permanent water, sewer, electricity and telecommunications connections.
- (b) MPAN and meter information.
- (c) Final meter readings.
- (d) Water-meter numbers and electric-vehicle charging meter information.
- (e) Utility-provider details.
- (f) Transfer of landlord accounts.
- (g) Export-tariff information.

6.6.2 Keys, access and controls

- (a) Apartment and communal keys.
- (b) Two clearly marked sets of apartment, communal, plantroom and service keys unless the Employer's Requirements state otherwise.
- (c) Window-lock and meter-cupboard keys.
- (d) Fobs and access codes.
- (e) Access-control information.
- (f) Fire-alarm-panel information.
- (g) Generator controls and emergency procedures.
- (h) Facilities-management training.

6.6.3 Resident information

- (a) Resident user guides.
- (b) Appliance manuals.
- (c) Manufacturer warranties and guarantees for all apartment appliances, sanitary fittings, heating equipment and renewable-energy equipment.
- (d) Local-amenity, utility-supplier and building-management information required for resident packs.
- (e) Heating, hot-water and ventilation instructions.
- (f) Fire-safety information.
- (g) Utility-supplier information.
- (h) Emergency contacts.
- (i) Building signage and wayfinding information.
- (j) Manufacturer warranties.

6.6.4 Estate and management-company handover

- (a) Formal handover of communal areas and external estate assets.
- (b) Communal asset register and estate, boundary and maintenance-responsibility plans.
- (c) Service contracts, maintenance agreements and landscape-establishment obligations.
- (d) External-lighting, drainage, gate, access-control, utility and landlord-account information.
- (e) Fire and life-safety management information and facilities-management training.
- (f) Records of keys, fobs, access codes, plant controls, outstanding defects and seasonal maintenance.

6.7 Post-Completion

6.7.1 Defects period

- (a) Defect reporting and investigation.
- (b) Defect rectification.

- (c) Access coordination.
- (d) Retesting.
- (e) Closure records.
- (f) End-of-defects inspection.

6.7.2 Commercial close-out

- (a) Subcontract final accounts.
- (b) Main-contract final account.
- (c) Variation close-out.
- (d) Retention management.
- (e) Commercial-record completion.

Annex A — Level 4 Zone Breakdown

Each applicable Level 3 deliverable is allocated to the relevant physical zone using the zone codes below. Zone allocation is a location overlay; it does not alter the decimal deliverable numbering.

Zone code	Description
Zone A	Apartments
Zone B	Apartments
Zone C	Apartments
Zone D	Apartments
Zone E	Apartments
Core	Central core and common circulation
Roof	Roof and roof-level plant
External	External works and site-wide infrastructure

Annex B — Level 5 Floor Breakdown

Each applicable zone is divided by floor or storey in accordance with the approved architectural general-arrangement drawings. Location codes take the form Zone.Floor. Only floors applicable to the relevant zone are used.

Example location code	Meaning
A.00	Zone A, ground floor
A.01	Zone A, first floor
A.02	Zone A, second floor
Core.00	Core, ground floor
Roof	Roof-level works
External	Site-wide external works

Location-coding protocol: apply the Zone.Floor code as a suffix to a Level 3 deliverable reference only where zone/floor resolution is required (for example 4.3.6.1 · A.00). Deliverables are not duplicated across every zone and floor; the overlay is applied selectively as the design develops.

PROJECT SCOPE EXCLUSIONS

Unless expressly identified as included within the Contract Documents, approved drawings, specifications or schedules, the following are excluded from the Works:

- (a) Loose furniture, fixtures and equipment.
- (b) Beds, mattresses, sofas, chairs and tables.
- (c) Loose reception and communal-area furniture.
- (d) Fitted wardrobes and internal wardrobe fittings.
- (e) Curtains, window blinds, curtain tracks and poles.
- (f) Mirrors and fixed bathroom accessories.
- (g) Toilet-roll holders, towel rings, shelves and decorative accessories.
- (h) Artwork, pictures, ornaments and decorative displays.
- (i) Television sets and loose audio-visual equipment.
- (j) Computers, printers, telephones and loose office equipment.
- (k) Loose kitchen equipment, crockery, utensils and small appliances.
- (l) Freestanding appliances not identified in the approved appliance schedule.
- (m) Proprietary parcel-locker systems unless identified in the approved parcel-store schedule.
- (n) Balcony, terrace and garden furniture.
- (o) Decorative planters and loose planting containers.
- (p) Show-apartment furniture, dressing and staging.
- (q) Marketing-suite furniture displays and sales equipment.
- (r) Marketing signage, sales boards and promotional branding.
- (s) Apartment upgrade packs and purchaser-selected options.
- (t) Purchaser variations issued after the agreed design freeze.
- (u) Resident possessions and personal equipment.
- (v) Management-company office equipment and operational supplies.
- (w) Cleaning machines, loose cleaning equipment and consumables.
- (x) Refuse bins and movable waste containers unless specifically scheduled.
- (y) Replacement filters, lamps, batteries and consumables beyond the initial handover provision.
- (z) Operational staffing, concierge services and facilities-management mobilisation.
- (aa) Utility consumption and service charges following handover.
- (bb) Maintenance and service contracts beyond the period expressly included within the Contract.
- (cc) Employer, funder, sales, marketing, legal and finance costs.
- (dd) Employer-supplied items not expressly identified within the relevant trade-package scope.
- (ee) Additional work resulting from purchaser, operator or management-company changes after approval of the coordinated design.